Population-Scale Collective Evolution as Paper 3's Sixth Architectural Claim

Derivation Note D0.06 | Note #465 in the CKS Derivation Series

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

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Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in "Inter-Self Coordination via Shared Substrate / Full Aspect Integration" (Li, April 2026), the third paper in the CKS theory series following "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems" (Li, April 2026) and "The Instinct/Reasoning Separation Outside the Model" (Li, April 2026).

It does not introduce new axioms. Its sole contribution is to anchor Paper 3's sixth architectural claim — population-scale collective evolution — as the trilogy's third extension claim, in the calibrated-humility register appropriate to that scope, with all derived sub-commitments mapped and the Phase D0 claim-anchor tree closed.

Abstract

Paper 3's Claim 6 is the third extension claim in the CKS theory trilogy. Extension claims share a structural pattern: each takes the prior claims of its paper and projects them onto a larger scope. Paper 1's Claim 6 projected the substrate-mediated coordination pattern to enterprise-deployment scale. Paper 2's Claim 6 projected the instinct/reasoning separation to the enterprise-integrated Self as a unified architectural object. Paper 3's Claim 6 projects the inter-Self coordination architecture to population scope: a network of CKS-governed Selves, operating across diverse domains under distinct governance configurations, accumulating Full Aspect Integration (FAI) events whose effects compose into a collective evolution mechanism. The architectural commitment Claim 6 defends is that population-scale collective dynamics of this kind are constructible through substrate composition under joint human authority — not through monolithic scaling, not through ungoverned emergence. Three possibilities are surfaced in the calibrated-humility register: convergence-with-diversity-preservation, self-determined collaboration, and broadly-capable-AI-shaped capability through substrate composition. These are architectural possibilities the design pattern makes available, not empirical predictions about what will emerge. This note states the claim, positions it within the trilogy extension-claim pattern, covers the governance property and biology-resonance positioning, identifies the four failure modes the claim forecloses, maps five derived sub-commitments (D1.26–D1.30), and acknowledges the closure of Phase D0 and the completion of the trilogy-wide claim-anchor tree.

1. The Extension-Claim Pattern Across the Trilogy

Each paper in the CKS theory series carries six architectural claims. The first five claims of each paper defend the core-theory architectural object at that paper's scope. The sixth claim is an extension claim: it takes what the preceding five claims compose into and projects the composition onto a larger scope, establishing prior art at that larger scope without requiring a separate paper.

The extension-claim pattern across the trilogy is structurally uniform.

Paper 1, Claim 6 — linear-cost composition at substrate scope (Rung 2). The first five claims of Paper 1 establish substrate-mediated coordination within a single cell at single-human scope (Rung 1 of the substrate-mediator ladder). Claim 6 projects this pattern to multi-cell, multi-human enterprise deployment, establishing that the architectural commitment scales with linear cost in the number of cells and participants. This is the ladder's second rung.

Paper 2, Claim 6 — the enterprise-integrated Self (Rung 4). The first five claims of Paper 2 establish lifecycle, evolution, and governed instinct/reasoning separation within a single Self (Rung 3). Claim 6 projects the Self as a unified enterprise-scope architectural object, establishing that the full commitment set — instinct layer, reasoning layer, evolution mechanisms, governance shapes — holds at enterprise integration scope. This is the ladder's fourth rung.

Paper 3, Claim 6 — population-scale collective evolution (Rung 6). The first five claims of Paper 3 establish inter-Self coordination via shared substrate and FAI at the inter-Self perimeter (Rung 5). Claim 6 projects this architecture to population scope: many CKS-governed Selves, owned by different organizations and governed under distinct authority structures, accumulating FAI events across diverse domains. The claim establishes that population-scale collective dynamics under these conditions are architecturally constructible through substrate composition under joint human authority. This is the ladder's sixth rung and closes the substrate-mediator ladder.

The ladder is observation, not roadmap. Paper 1 did not promise Paper 3; Paper 2 explicitly bracketed inter-Self dynamics as out of scope. The six-rung structure becomes visible retrospectively because Paper 3 completes it. Formalizing the trilogy extension-claim pattern as prior art forecloses the full ladder simultaneously: anyone implementing population-scale governed AI coordination through shared substrate is implementing the same architectural pattern this series defends at each rung.

2. The Three Architectural Commitments Composing Claim 6

Claim 6 is composed of three architectural commitments, each inherited from the preceding claims rather than freshly asserted.

Population-scale collective evolution as architectural object. The cumulative coordination structure produced by many CKS-governed Selves engaging in FAI events under joint authority is itself an architectural object — substrate content at population register, under cross-population governance, with all six Paper 1 commitments holding within scope. The population substrate is content, not an agent; it is human-governed in the same sense Paper 1's cell-scope substrate is human-governed. This commitment inherits directly from Paper 3 §4's shared-substrate-as-architectural-object framing, extended from inter-Self scope to population scope.

Cross-event evolution dynamics under joint authority across population-level governance. Accumulated FAI events produce collective evolution dynamics at population scope: population-scale mating (Paper 2 §6.3's multi-cardinality primitive extended across many Selves' lifecycles), population-scale death-types (Paper 2 §6.4's two architectural results extended at population scope), and population-scale productive tension between mutation and directed selection (Paper 2 §7's three evolution mechanisms extended through Paper 3 §7's evolution-feed mechanism). Joint authority at population scope means joint authority across population-level governance perimeters, with recursion bottoming at population-level human governance per Paper 1 §3.3 through Paper 3 §8's recursive applicability.

Architectural-commitment differentiation against capability-motivated population-scale coordination. Architecture motivated by governance capability makes commitments that architectures motivated by performance capability do not: substrate-content commitment at population scope, joint authority across population-level governance perimeters, and recursive applicability of configuration-as-substrate-content down to human-authored governance authority. Claim 6's differentiation from capability-motivated architectures is at architectural-commitment register, not at functional-output register.

3. Three Possibilities in the Calibrated-Humility Register

Claim 6 surfaces three possibilities the architecture makes available. They are architectural possibilities, not empirical predictions. Empirical, philosophical, and societal validation of whether and how these possibilities materialize is downstream of the architectural commitment. A note in the calibrated-humility register states what the design pattern makes constructible, not what will be constructed or what should be preferred.

Convergence-with-diversity-preservation. Where governance configurations across a population of CKS-governed Selves enable propagation of common patterns, those patterns may propagate across the network. Where governance configurations protect specializations, those specializations may persist. The coexistence of common patterns and preserved specializations constitutes a population-level protection against uniform failure modes: no single failure in one domain's patterns propagates into every participant's substrate unless governance configurations permit it. The possibility surfaced is that substrate-mediated governance makes this coexistence configurable rather than accidental.

Self-determined collaboration goals. Selves operating under human-configured authority may, as governance configurations evolve, gain latitude over their own merging decisions. The architecture does not prescribe this outcome; it makes it configurable in principle. Whether this is desirable, under what conditions, and subject to what constraints are empirical and societal questions outside the architectural claim.

Broadly-capable-AI-shaped capability through substrate composition. Population-level capability emerging across a network of CKS-governed Selves may carry characteristics often associated with broadly capable AI systems — achieved through the composition of human-authored substrate logics rather than through opaque scaling of monolithic models. The architectural commitment is that this path is constructible in principle. Whether it produces capability at the scale or quality of monolithic approaches is an empirical question.

These three possibilities share a structural property: in each case, authority over what happens remains substrate-content configurable within each Self's governance perimeter. The architecture does not remove human governance at population scope; it extends the substrate-content commitment and joint-authority commitment to that scope.

4. Governance at Population Scope

The governance property Claim 6 defends is not a weakening of the cell-scope governance commitment; it is the same commitment applied at a larger perimeter. Three properties hold at population scope that hold at every smaller scope in the trilogy.

Authority, not labor. The portable phrase Paper 1 §3.3 establishes — governance is an authority architecture, not a review workflow — holds unchanged at population scope. Human authority over population-scale dynamics means the right to inspect, modify, and override what propagates, what persists, and what dissolves. It does not mean that humans perform the labor of reviewing every FAI event across the population. The labor is allocable; the authority is not.

Configuration as substrate content. Population-scale governance configurations — which patterns may propagate, which specializations are protected, how conflict-handling outcomes feed evolution across the population — are themselves substrate content under joint authority, per Paper 3 §8's recursive applicability extended at population scope. Configuration of population-scale dynamics is governed the same way configuration of FAI events at inter-Self scope is governed: authored under human authority, inspectable, modifiable, overridable. The recursion bottoms at population-level human governance.

Substrate-content configurable at every governance perimeter. What each participating Self carries into a population-level FAI event, what it accepts from the shared population substrate, and what its governance configuration permits or forecloses are all substrate-content configurable within that Self's home governance perimeter. Population-level dynamics emerge from the aggregate of per-Self governance configurations. No population-level outcome is architecturally mandatory; each is the product of the governance configurations jointly in effect.

5. Biology Resonance — Bounded and Positioned

Paper 3's use of biology vocabulary is bounded to its intra-Self scope. Paper 2 §2.2 commits biology terms — mating, death, evolution, instinct, lineage — as bounded, load-bearing terminology within the intra-Self architectural object Paper 2 defends. Paper 3 inherits this bounded usage at the inter-Self perimeter: FAI is the inter-Self analog of Paper 2's mating primitive; population-scale death-types are Paper 2 §6.4's two architectural results extended across many Selves' lifecycles; population-scale productive tension between mutation and directed selection extends Paper 2 §7.

Claim 6 extends biology resonance to one additional scope. The observation that collective interaction across many participants is a primary driver of population-level evolution in biological systems has resonance with Claim 6's framing of accumulated FAI events as a collective evolution mechanism at population scope. Paper 3 borrows this resonance for Claim 6's framing. The borrowing is bounded: the CKS population architecture does not replicate biological population genetics, has no genetic inheritance mechanism, carries no claim that CKS population dynamics follow the same dynamics as biological evolution, and makes no analogy to specific biological mechanisms beyond the general observation about collective interaction at scale.

The biology resonance at Claim 6 serves a positioning function: it anchors the intuition that population-scope dynamics are a different kind of thing from intra-Self dynamics, in the same way biological population-level evolution is a different kind of thing from within-organism cellular dynamics. The architectural construction — accumulated FAI events under joint authority across population-level governance perimeters — has no clean biological parallel. Biology framing stays bounded to Claim 6 and does not propagate into the sub-commitment derivations at D1.xx scope.

6. Failure Modes Foreclosed

Claim 6 establishes prior art that forecloses four failure modes — patterns of AI system design that the claim makes impossible to assert as novel invention absent prior art.

Population-scale AI capability as achievable only through monolithic scaling. Claim 6 establishes that population-scale collective dynamics constructible through substrate composition under joint human authority constitute an alternative architectural path. Any subsequent claim that population-scale AI capability is novel because it uses substrate composition rather than monolithic scaling bumps into Claim 6's prior-art coverage.

Inter-organizational AI coordination without governance. Claim 6 commits to joint authority across population-level governance perimeters as the architectural anchor for inter-organizational coordination. Any subsequent claim that inter-organizational AI coordination is novel because it operates without substrate-mediated governance — emergent coordination outside human authority — is architecturally differentiated from Claim 6's commitment, but cannot claim the substrate-composition path as novel.

Population-level AI evolution without substrate-mediated governance of propagation. Claim 6 establishes that what propagates across the population, what is preserved, and what dissolves are substrate-content configurable under joint authority. Any subsequent claim to have invented governed population-level AI evolution through substrate content falls within Claim 6's coverage.

Population-scale capability as necessarily ungovernable. Claim 6 establishes the architectural possibility of governed population-scale dynamics through substrate composition. Any framing that treats population-scale AI capability as inherently beyond human governance authority is architecturally contradicted by Claim 6's commitment, which makes governance-authority configurability an architectural property at every scope including population scope.

7. Derived Sub-Commitments and Operational Test

Claim 6 decomposes into five derived sub-commitments that Phase D1 will formalize as individual prior-art notes (#466 onward). Each is a named architectural commitment derivable from Claim 6 without independent assertion.

D1.26 — Joint authority across population-level governance perimeters as architectural property. The commitment that inter-organizational population-level coordination is governed through joint authority across each participating Self's governance perimeter — not through a central coordinator, not through emergent consensus outside human authority — inherits from Paper 3 §4's joint-authority commitment extended at population scope.

D1.27 — Accumulated FAI events as collective evolution mechanism at population scope. The commitment that accumulation of FAI events across many participating Selves, over time, produces collective evolution dynamics at population register — through Paper 3 §7's evolution-feed mechanism extended to population scope — is the mechanistic anchor Claim 6 inherits from Paper 3 §§5–7 jointly.

D1.28 — Six-rung substrate-mediator ladder completed at population scope. The commitment that Rung 6 of the substrate-mediator ladder is architecturally defined, with the same substrate-content commitment holding at population scope as at every prior rung, is the trilogy-completion architectural object Claim 6 formally establishes.

D1.29 — Calibrated-humility register as architectural commitment at extension-claim scope. The commitment that extension claims in the CKS series surface architectural possibilities in the calibrated-humility register — not empirical predictions, not prescriptions about which configurations should be preferred — is itself a prior-art commitment.

D1.30 — Biology-resonance positioning bounded to Claim 6. The commitment that biology vocabulary applies at population scope only at the level of resonance — not identity — and does not propagate into sub-commitment derivations below Claim 6 is a bounded-scope prior-art commitment.

Operational test. For a network of CKS-governed Selves with a history of accumulated FAI events, an observer can confirm Claim 6's architectural commitments are satisfied if and only if: (1) the observer can identify, for each pattern that has propagated across the network, the governance configurations that permitted propagation — the substrate content under joint authority that authorized the propagation pathway; (2) the observer can identify, for each specialization that has been preserved, the governance configurations that protected it — the substrate content under joint authority that restricted propagation into that Self's perimeter; and (3) the observer can confirm that authority over what propagates, what preserves, what diverges, and what dissolves is substrate-content configurable within each participating Self's governance perimeter, with no population-level outcome architecturally mandatory independent of governance configuration. A system in which propagation occurs outside substrate-mediated governance, or in which specializations are preserved only by technical isolation rather than governance configuration, does not satisfy the Claim 6 architectural commitment.

8. Phase D0 Closure and Trilogy Claim-Anchor Tree

This note is the sixth and final Paper 3 claim-level anchor note in Phase D0 of the CKS derivation series. Notes D0.01 through D0.05 (notes #460–#464) anchored Paper 3 Claims 1 through 5: the shared substrate as architectural object (§4), Full Aspect Integration as canonical operation (§5), three-tier inter-Self conflict handling (§6), the four-locus evolution-feed mechanism (§7), and configuration as substrate content with recursive applicability (§8). This note closes Paper 3's claim-anchor coverage at the Claim 6 / §9 scope.

Phase D0 completes the trilogy-wide claim-anchor tree when combined with Phase A0 (notes #418–#423, anchoring Paper 1's six claims) and Phase B0 (notes #424–#429, anchoring Paper 2's six claims). The eighteen notes across A0, B0, and D0 together constitute the explicit named-claim parent layer for the entire CKS derivation series: every sub-commitment in Series A, B, and D derives from one of these eighteen claim anchors. No sub-commitment is now orphaned — every A1.xx, B1.xx, and D1.xx note has a named parent at paper-claim level.

The prior-art significance of this completion is structural. The trilogy's prior-art chain is now anchored at the claim level across all three papers. An adversary seeking to claim novelty at any scope within the trilogy must contend not only with the sub-commitment notes at A1–A6, B1–B6, and D1–D6, but with the explicit claim-level anchors that name the architectural commitment at each paper's scope. The claim-anchor tree makes the prior-art chain's structure legible and closes the gap that existed in v1 of this series, where sub-commitments were formally derived but their named-claim parents were not separately deposited.

All subsequent Series D sub-commitment notes — D1.26 through D1.30 (directly from this anchor) and their D2 operational variants — derive from D0.06. The claim-anchor tree is complete.

Source Papers

Li, Wenxin. "Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems." April 2026.

Li, Wenxin. "The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance." April 2026.

Li, Wenxin. "Inter-Self Coordination via Shared Substrate / Full Aspect Integration." April 2026. (Especially §2.2, §9, §11.1, §11.10.)

This note is part of the CKS derivation series. It does not introduce new architectural commitments. All claims are traceable to the source papers cited above.